Happiest Health
Digital Twin Challenge 2026

Reimagining and Reforming Healthcare in India Summit 2026
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GlucoTwin

A Personalized, Continuously Updating Digital Twin for Early Prediction of Glucose Excursions

From Reactive Monitoring to **Proactive Care**

Historical Health Records
Demographics, labs, diagnoses, medications

+

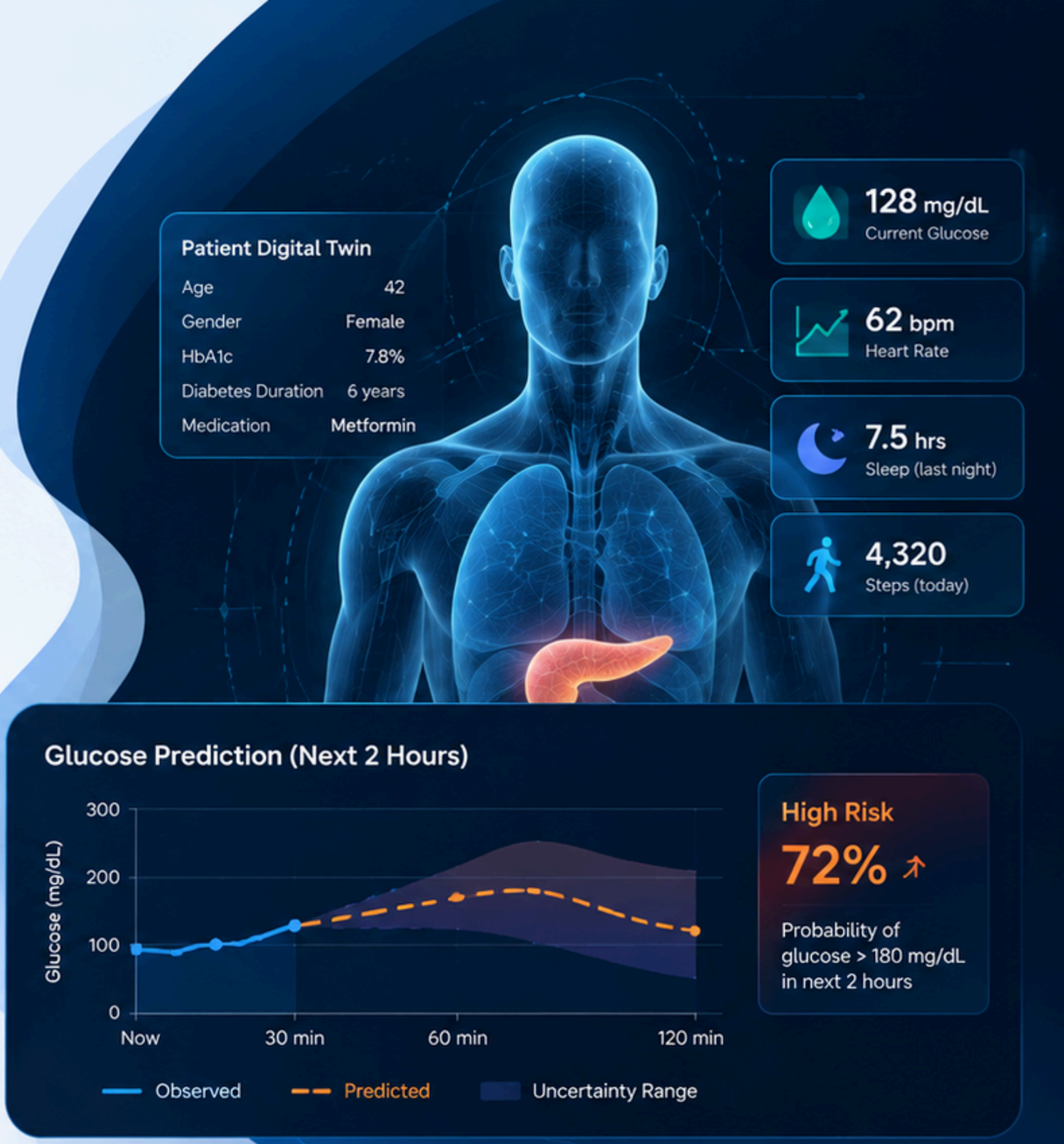
Wearable Sensor Data
Glucose, heart rate, sleep, activity

+

AI/ML Prediction
Early risk estimation (2-hour horizon)

+

Clinical Dashboard
Interactive patient digital twin



Team

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THE PROBLEM

Reactive Monitoring Is Not Enough

Type 2 Diabetes is a growing public health challenge in India. Traditional monitoring relies on **periodic clinical visits** and self-reported measurements, which often miss short-term glucose fluctuations that can lead to adverse health events.

101 million
people with diabetes in India (2023)
IDF Diabetes Atlas, 10th edition

136 million
estimated by 2045
IDF Diabetes Atlas, 10th edition

~50%
remain undiagnosed
IDF Diabetes Atlas, 10th edition

Key Challenges in Current Care

Periodic Monitoring
Clinical assessments and HbA1c tests provide only a snapshot in time.

Missed Fluctuations
Short-term glucose spikes and drops often go undetected between visits.

Lack of Personalization
Population-level guidance does not capture individual responses to meals, activity and lifestyle.

Higher Risk of Complications
Uncontrolled glucose variability increases the risk of cardiovascular disease, kidney damage and other long-term complications.

OUR INSIGHT

Continuous, personalized prediction of glucose excursions can enable **proactive monitoring and timely intervention**, reducing the risk of long-term complications.

OUR SOLUTION

A Personalized Digital Twin for Proactive Care

GlucoTwin creates a virtual, continuously updating representation of a patient by combining historical health records with real-time wearable data to predict glucose excursions **up to 2 hours in advance** and provide actionable insights for clinicians.

Historical Health Records
Demographics, lab results, diagnoses, medications

+

Real-time Wearable Data
Glucose, heart rate, sleep, steps, activity, meals

+

AI/ML Prediction
Early estimation of glucose excursions (2-hour horizon)

+

Clinical Dashboard
Interactive virtual patient for informed decisions

From static monitoring to a **dynamic, predictive** and **personalized** view of the patient's glucose health.

Early Risk Estimation
Predicts glucose excursions up to 2 hours in advance

Personalized Insights
Patient-specific modeling using combined data

Explainable Predictions
Shows key factors influencing risk

Supports Clinical Decisions
Helps healthcare providers identify high-risk patterns early

SYSTEM ARCHITECTURE

End-to-End Pipeline
From Data to Actionable Insights

A unified pipeline that fuses historical health records with real-time wearable data to maintain a **continuously updating** patient Digital Twin and predict glucose excursions **up to 2 hours in advance**.

1. Data Sources

Historical Health Records (Synthetic EHR)
- Demographics
- Diagnoses
- Lab results (HbA1c, fasting glucose)
- Medications
- Medical history

Wearable / Sensor Data (Synthetic Time Series)
- Continuous glucose (CGM)
- Heart rate & HRV
- Steps & activity
- Sleep data
- Meal and activity events

2. Data Preprocessing & Feature Engineering

Data Cleaning
- Missing value handling
- Time alignment
- Outlier treatment

Feature Engineering
- Glucose trends & variability
- Rolling statistics
- Activity & sleep features
- Meal-related features
- Patient-specific features

Train-Test Split
- Patient-level split
- Prevent data leakage

3. Patient Digital Twin (Real-time State Update)

Dynamic Patient State
- Current sensor readings
- Recent history (last 15/30/60 mins)
- Rolling statistics
- Baseline health profile
- Data quality indicators
- Personalized patient

Continuously Updated as New Data Arrive
- Real-time stream simulation
- State update at each timestep

4. Prediction Engine

AI/ML Model
- XGBoost (classification)
- Baseline model for comparison
- 2-hour prediction horizon
- Patient-specific features
- Feature importance (SHAP)

Output
- Probability of glucose > threshold in next 2 hours
- Predicted glucose trajectory (optional)
- Risk category (Low / Moderate / High)
- Key contributing factors

5. Clinical Dashboard (Streamlit Application)

- **Executive Overview** — Real-time status & risk
- **Patient Profile** — Historical health records
- **Live Digital Twin** — Glucose trends & predictions
- **Risk & Explainability** — Key factors influencing risk
- **Scenario Simulator** — What-if analysis (meals, activity)
- **Model Performance** — Evaluation metrics & comparison
- **Prediction History** — Past predictions and outcomes

KEY COMPONENTS

Data Fusion
Combines historical EHR data with real-time wearable data

Continuously Updating Twin
Maintains a dynamic, personalized patient state

Early Risk Prediction
Predicts glucose excursions up to 2 hours in advance

Clinician-Focused UI
Interactive dashboard for informed decision-making

DATA SOURCES & FUSION

Combining Historical Health Records with Real-time Sensor Data

GlucoTwin integrates static patient information with dynamic, time-series wearable data to create a comprehensive and personalized view of a patient's glucose health.

Synthetic Data Generation

- Generated realistic, anonymized patient data with clinically plausible distributions.
- Simulated glucose, heart rate, steps, sleep and meal patterns with temporal dependencies.
- Introduced inter-patient variability, sensor noise and occasional missing values.
- Fixed random seed for reproducibility.

1 Historical Health Records

Demographics, diagnoses, lab results and medical history (form synthetic EHR)

Demographics
Age, gender, BMI

Clinical History
Diagnoses, duration

Laboratory Results
HbA1c, fasting glucose

Medications
Type, dosage

Other Information
Lifestyle factors, etc.

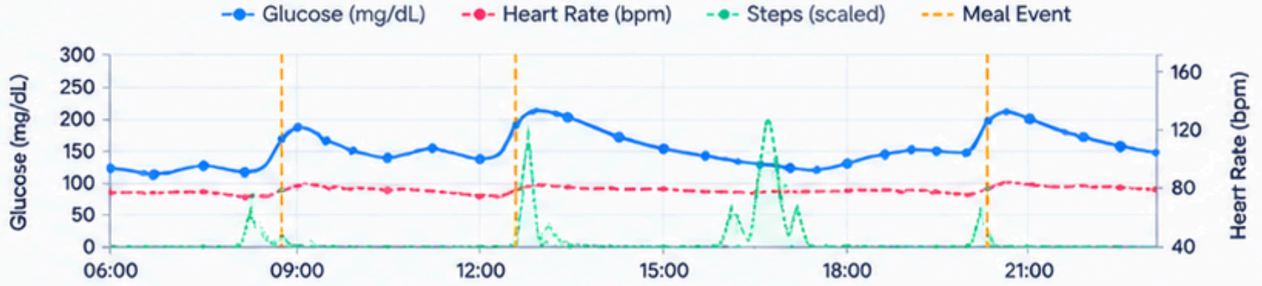
Sample Patient Record (Synthetic)

Patient ID	PT-0119
Age	42
Gender	Female
BMI (kg/m²)	27.4
Diabetes Duration (years)	6
HbA1c (%)	7.8
Fasting Glucose (mg/dL)	142
Diagnoses	Type 2 Diabetes, Hypertension
Medications	Metformin
Family History	Diabetes (Yes)
Smoking	No
Exercise Frequency	2–3 times/week

2 Real-time Wearable Sensor Data

Continuous glucose monitoring and associated physiological signals

Sample Time-series Data (PT-0119)



Sample Sensor Data (5-minute interval)

Timestamp	Glucose (mg/dL)	Heart Rate (bpm)	HRV (ms)	Steps	Sleep (hrs)	Meal Event	Activity
2026-10-15 11:55	118	78	42	120	-	No	Rest
2026-10-15 12:00	125	82	40	210	-	Yes	Meal
2026-10-15 12:05	142	88	38	320	-	No	Rest
2026-10-15 12:10	168	92	36	410	-	No	Rest
2026-10-15 12:15	182	94	34	380	-	No	Light Activity

3 Data Fusion

Aligning and combining multi-source data to create a unified patient representation

Historical EHR (static features) + **Wearable Sensors** (time-series features) → **Preprocessing & Feature Engineering** → **Unified Patient State** (for ML prediction)

THE DIGITAL TWIN ENGINE

A Continuously Updating Virtual Patient

GlucoTwin maintains a dynamic, patient-specific virtual state that evolves as new sensor observations arrive, combining historical context with real-time data to generate personalized predictions.

1. **Initialize Patient Twin**
   Load historical health records and baseline statistics
2. **Ingest New Sensor Data**
   Update state with latest glucose, heart rate, activity, sleep and meal events
3. **Update Features in Real Time**
   Compute rolling metrics, trends and contextual features
4. **Generate Prediction**
   Estimate probability of glucose excursion in the next 2 hours
5. **Store and Visualize**
   Log prediction, update dashboard and maintain prediction history

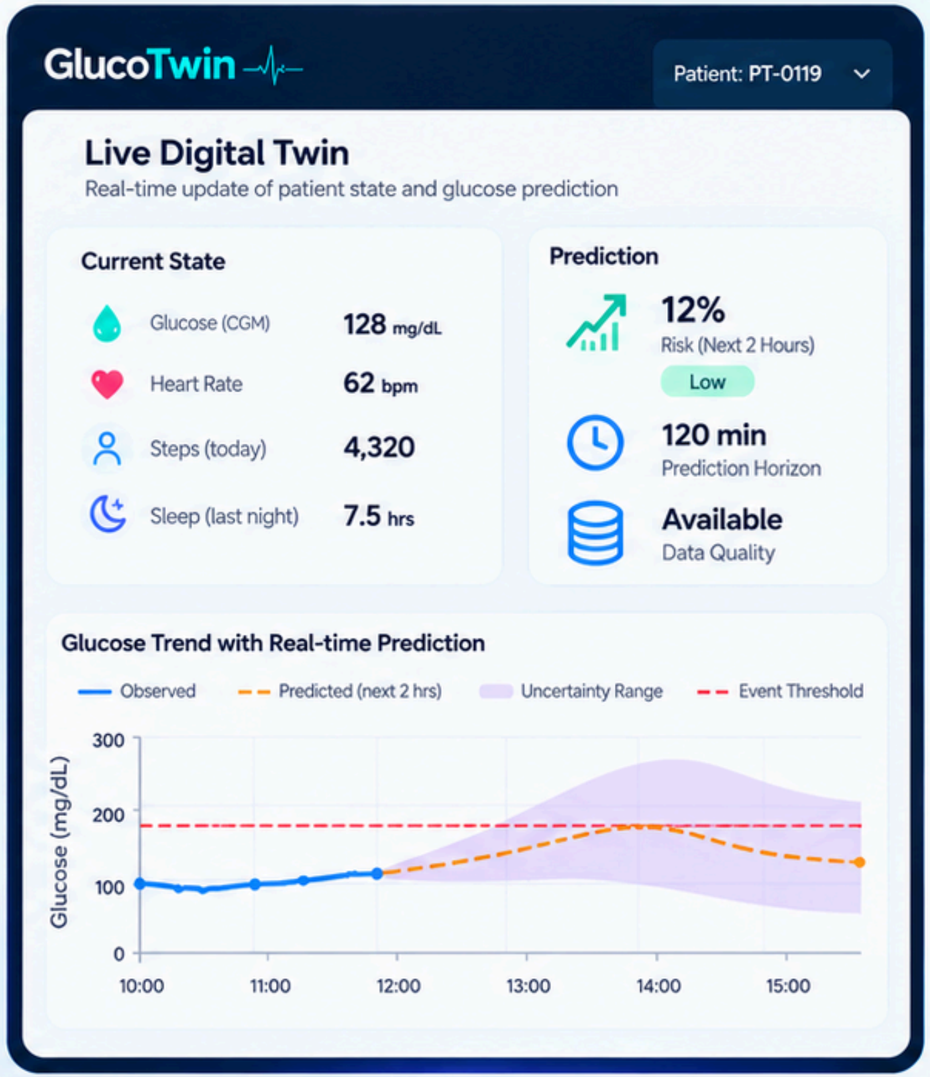
Digital Twin State

Historical Profile
Demographics
Lab values
Diagnoses
Medications

Real-time Sensors
Glucose (CGM)
Heart rate
Steps & activity
Sleep data
Meal events

Derived Features
Rolling glucose stats
Glucose variability
Recent trend (slope)
Time since meal
Activity & sleep features

Updated Twin State
Current health status
Dynamic feature set
Prediction inputs
Data quality indicators



KEY CAPABILITIES

Real-time State Updates
Twin state evolves with every new observation

Patient-Specific Modeling
Uses individual history and dynamic features

2-Hour Risk Prediction
Estimates probability of glucose excursion ahead

Prediction History
Stores predictions for retrospective analysis

AI/ML METHODOLOGY

Predicting Glucose Excursions Using Patient-Specific Modeling

We design a machine learning pipeline that leverages engineered features from historical records and real-time sensor data to predict the probability of a glucose excursion in the **next 2 hours** for each individual patient.

Prediction Task

Estimate the probability of a glucose excursion (hyperglycemia or hypoglycemia) within the next 2 hours, updated continuously as new data arrives.

Why 2 Hours?

- Early warning enables timely and proactive intervention
- Captures short-term meal and activity effects
- Clinically relevant prediction horizon for day-to-day care

1 Feature Engineering

- **Glucose trend features** (recent values, slope, variability)
- **Physiological features** (heart rate, HRV, steps, sleep)
- **Meal-related features** (meal timing, carbs, meal events)
- **Historical features** (demographics, diagnoses, labs)
- **Time-based features** (time of day, day of week)
- **Patient-specific features** (baseline levels, personal patterns)

2 Model Architecture

Gradient Boosted Decision Trees (XGBoost)

- Handles non-linear relationships
- Robust to heterogeneous features
- Provides feature importance for interpretability
- Efficient and well-suited for tabular + time-series features

Prediction
(Probability of Glucose Excursion)

3 Training & Evaluation Setup

- **Patient-level train-test split** — Prevents data leakage across patients
- **Time-aware split** — Maintains temporal order of data
- **Data preprocessing** — Missing value handling, normalization, feature scaling
- **Class imbalance handling** — Weighted loss / resampling
- **Evaluation metrics** — Precision, Recall, F1-score, PR-AUC

4 Prediction Output

Probability of Excursion
(in next 2 hours)

Prediction Types

- **Hyperglycemia** — Glucose > 180 mg/dL
- **Hypoglycemia** — Glucose < 70 mg/dL

Example Prediction Pipeline

MODEL STRENGTHS

- Accurate short-term prediction of glucose excursions
- Patient-specific personalization
- Interpretable predictions with feature importance (SHAP)
- Robust to noisy and missing real-world sensor data

EVALUATION RESULTS

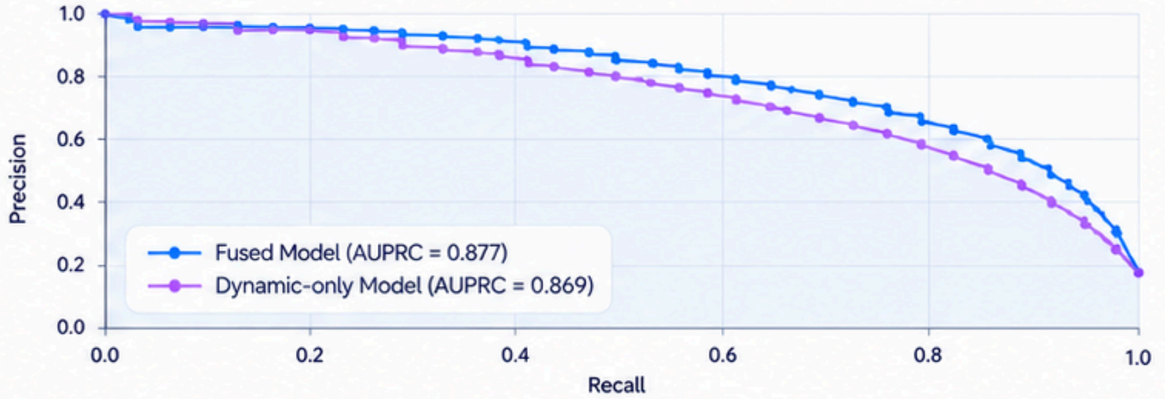
Model Performance on Held-out Patients

We evaluate our models on unseen synthetic-patient windows and compare a multimodal (fused) model with a dynamic-only model to predict glucose excursions within the **next 2 hours**.

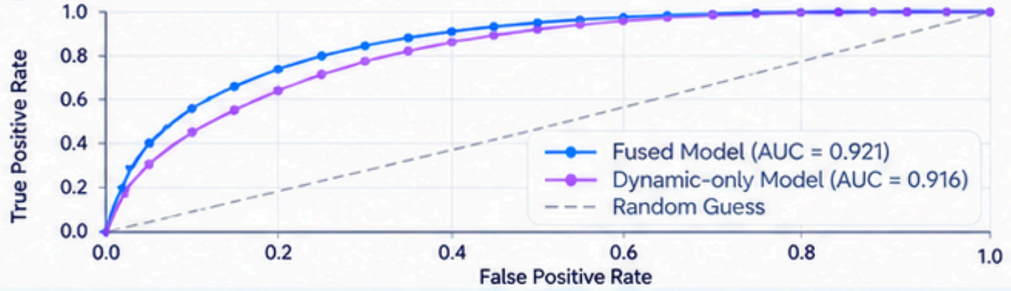
1 Overall Performance (Held-out Test Set)

	Fused Model (Our Approach)	Dynamic-only Model
Precision Correctly identified excursions	0.873	0.865
Recall Captured high-risk events	0.819	0.830
F1-Score Balanced performance	0.845	0.847

2 Precision–Recall Curve



3 Receiver Operating Characteristic (ROC) Curve



4 Performance Across Glucose Excursion Types

Event Type	Fused Model			Dynamic-only Model		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Hyperglycemia (Glucose > 180 mg/dL)	0.861	0.803	0.831	0.852	0.811	0.831
Hypoglycemia (Glucose < 70 mg/dL)	0.885	0.835	0.859	0.878	0.849	0.863

KEY INSIGHTS

- **High discrimination** — AUC = 0.921 on unseen patients
- **Balanced performance** — F1 = 0.845 for multimodal model
- **Robust across event types** — Consistent performance for hyperglycemia and hypoglycemia

REAL-TIME DEMONSTRATION

From Sensor Data to Actionable Insights

Our system continuously ingests real-time CGM data, predicts future glucose trends for the **next 2 hours**, and provides early warnings for potential hyperglycemia or hypoglycemia events.

1 Real-time Input

Live CGM data and contextual information are fed into the digital twin.

CGM Readings
Glucose value every 5 minutes

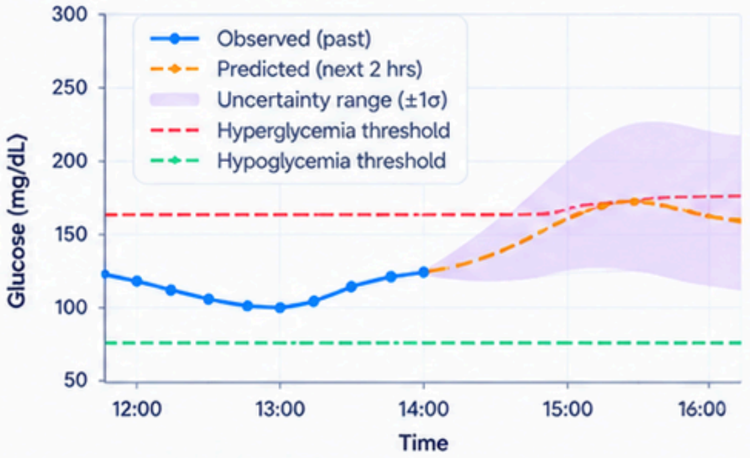
Meal Information
Time, type and estimated carbs

Physical Activity
Activity type and intensity

Patient Profile
Age, BMI, medical history

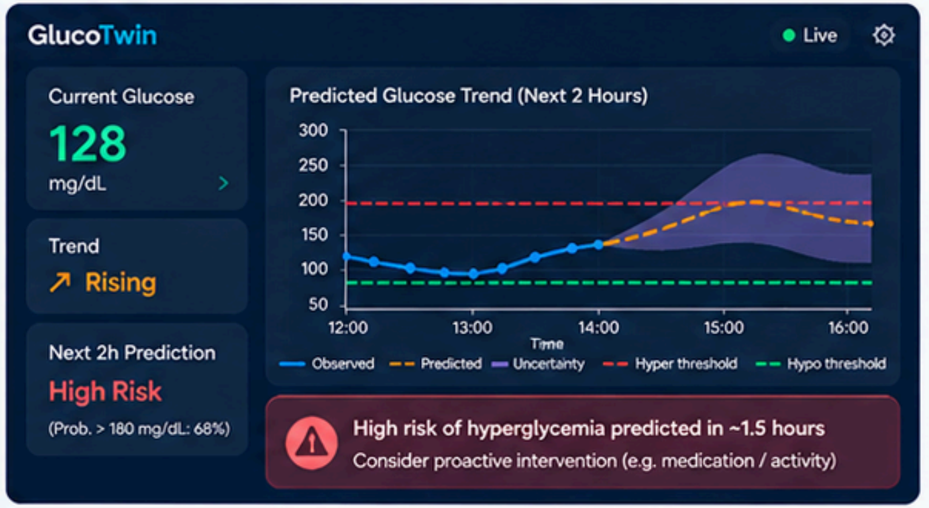
2 Real-time Prediction

The model predicts glucose levels for the next 2 hours and quantifies uncertainty.



3 User Interface (Demo)

Interactive dashboard showing real-time readings, predictions and alerts.



4 Example Scenarios

The digital twin adapts to individual patterns and provides early warnings.

Post-Meal Hyperglycemia Risk
After a high-carb meal, the model predicts a potential glucose spike ~1.5 hours in advance, enabling timely intervention.

Exercise-Induced Hypoglycemia
During and after physical activity, the model predicts a potential dip and issues an early warning to prevent low glucose events.

5 Key Benefits

Early Prediction
Up to 2 hours in advance

Personalized
Adapts to individual patient patterns

Proactive Care
Enables timely interventions

Real-time
Works with continuous CGM data

IMPACT AND FUTURE WORK

Towards a Safer, More Personalized Future

GlucoTwin demonstrates the potential of personalized digital twins for early prediction of glucose excursions, with applications in proactive diabetes management and continuous, data-driven care.

1 Potential Impact

Early Risk Detection
Predicts potential hyperglycemia and hypoglycemia up to 2 hours in advance, enabling timely intervention.

Personalized Care
Adapts to individual patterns using a personalized digital twin, leading to more relevant and accurate insights.

Informed Decision-Making
Provides actionable predictions and uncertainty estimates to support patients and healthcare providers.

Scalable and Real-world Ready
Designed to integrate with real CGM devices and can be extended for larger, more diverse populations.

2 Future Enhancements

Real CGM Integration
Deploy with real-time CGM devices (e.g., Dexcom, FreeStyle Libre) and validate on real patient data.

Richer Multi-modal Inputs
Incorporate additional signals such as sleep patterns, stress levels and heart rate for improved predictions.

Advanced Modeling Techniques
Explore transformer-based models and longer temporal context to capture complex patterns.

Clinical Validation
Collaborate with healthcare professionals to validate the model in real-world clinical settings.

3 Long-Term Vision

A comprehensive, personalized digital health companion for better diabetes management and improved quality of life.

2026 — **Real-time Deployment**
Integrate with real CGM devices and deploy for live, real-time predictions.

2027 — **Clinical Studies**
Validate with larger, diverse patient populations in clinical settings.

2028 — **Personalized Health Platform**
Expand to a holistic digital health companion with multi-modal data and personalized insights.

2030+ — **Broader Impact**
Enable scalable, preventive diabetes care to improve health outcomes and quality of life. globally.

KEY TAKEAWAYS

- Early prediction of glucose excursions up to 2 hours in advance.
- Personalized, adaptive modeling using a digital twin approach.
- Potential for real-world impact through integration, validation and scale.

CONCLUSION

Key Takeaways and Closing Thoughts

GlucoTwin shows that a personalized, continuously updating digital twin can effectively predict glucose excursions and provide early, actionable insights for better diabetes management.

1

Effective Prediction

Our multimodal (fused) model achieves strong performance (Precision: 0.873, Recall: 0.819, F1: 0.845) on held-out patients, accurately identifying high-risk glucose events up to 2 hours in advance.

2

Personalized and Adaptive

The digital twin adapts to individual patient patterns using multi-modal data, enabling more relevant and accurate predictions than a generic, one-size-fits-all approach.

3

Real-time and Actionable

Continuous ingestion of CGM data and contextual information allows for real-time prediction and early warnings, supporting timely interventions for hyperglycemia and hypoglycemia.

4

Scalable and Real-world Potential

The system is designed to integrate with real CGM devices and can be extended for larger, more diverse patient populations, paving the way for real-world deployment in diabetes care.

Overall Impact

GlucoTwin has the potential to **empower individuals with diabetes**, support proactive care, and contribute to **better health outcomes and quality of life**.

Future Directions

Larger Real-world Validation
Evaluate on real CGM datasets with diverse patient populations.

Integration with CGM Devices
Deploy with real-time data from devices like Dexcom and FreeStyle Libre.

Advanced Modeling
Explore transformer-based models and longer temporal context for improved predictions.

Holistic Health Platform
Extend to include additional physiological and lifestyle signals for more comprehensive insights.

FINAL MESSAGE

Towards a future of proactive, personalized diabetes care.

From data to insights, from insights to healthier lives.

GlucoTwin
A step towards a safer, healthier tomorrow.

Thank You

We appreciate the opportunity to present GlucoTwin. We hope our work contributes towards a future of **proactive**, **personalized** and **accessible** diabetes care.

Early Prediction
Up to 2 hours in advance

Personalized Insights
Adapts to individual patterns

Proactive Care
Supports timely interventions

Real-world Potential
Scalable and integrable with CGM devices

LET'S CONNECT

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